

ADVANCED MICROELECTRONICS, VLSI & INDUSTRIAL IOT

New Egyptian Baccalaureate (EG-Bac) Microelectronics & Industrial IoT Framework (Grade 12)

ACADEMIC YEAR: 2024 / 2025 | GRADE 12 ADVANCED

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Advanced Egyptian Baccalaureate textbook in Microelectronics, VLSI Design & Industrial IoT: 3D FinFET short-channel electrostatics, CMOS inverter noise margins and clock gating, FPGA 6-LUT architectures and Static Timing Analysis (STA), Q15 fixed-point DSP filtering, DMA circular streaming, Modbus RTU, Time-Sensitive Networking (TSN), on-device TinyML neural network inference, and silicon Root of Trust (RoT) with Physically Unclonable Functions (PUF).

CORE CURRICULUM HIGHLIGHTS:

- Nanometer FinFET DIBL mitigation, subthreshold swing, and layout design rule checking (DRC/LVS)
- Verilog pipelined datapath design, hazard bypassing, and setup/hold static timing slack closure
- DVFS low-power optimization, fixed-point Q15 fractional arithmetic, and zero-CPU DMA transfers
- Modbus RTU framing, IEEE 802.1Qbv TSN scheduling, INT8 TinyML quantization, and secure boot RoT

APPROVED CHAPTER MODULES:

- Nanometer CMOS VLSI, FinFET Physics & Integrated Circuit Layout	pp. 1 78 (3 core units)
- FPGA Architecture, HDL Synthesis & Static Timing Analysis	pp. 79 156 (3 core units)
- Low-Power Edge Computing, DSP & Embedded Telemetry	pp. 157 232 (3 core units)
- Industrial IoT (IIoT), Edge AI & Hardware Security	pp. 233 310 (3 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official EG-Bac Microelectronics, VLSI & Industrial IoT Compendium (8p - 34.
Full Ministerial Textbook: 310 Pages - 39.2 MB (Direct on MOE Portal)
Publisher: Egyptian Baccalaureate Board & Microelectronics Curriculum Authority
Official Portal: <https://moe.gov.eg/ar/elearning-content/> - Code: EGBAC-ELEC-2025
Legal Deposit: 2025/14892 - ISBN: 978-977-12-5543-87



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

Advanced Egyptian Baccalaureate textbook in Microelectronics, VLSI Design & Industrial IoT: 3D FinFET short-channel electrostatics, CMOS inverter noise margins and clock gating, FPGA 6-LUT architectures and Static Timing Analysis (STA), Q15 fixed-point DSP filtering, DMA circular streaming, Modbus RTU, Time-Sensitive Networking (TSN), on-device TinyML neural network inference, and silicon Root of Trust (RoT) with Physically Unclonable Functions (PUF). Nanometer FinFET DIBL mitigation, subthreshold swing, and layout design rule checking (DRC/LVS) Verilog pipelined datapath design, hazard bypassing, and setup/hold static timing slack closure DVFS low-power optimization, fixed-point Q15 fractional arithmetic, and zero-CPU DMA transfers Modbus RTU framing, IEEE 802.1Qbv TSN scheduling, INT8 TinyML quantization, and secure boot RoT

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Nanometer CMOS VLSI, FinFET Physics & Integrated Circuit Layout	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
FPGA Architecture, HDL Synthesis & Static Timing Analysis	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Low-Power Edge Computing, DSP & Embedded Telemetry	7 wks	15 pts	25.0%	3 core instructional units covering national standards and
Industrial IoT (IIoT), Edge AI & Hardware Security	7 wks	15 pts	25.0%	3 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: NANOMETER CMOS VLSI, FINFET PHYSICS & INTEGRATED CIRCUIT LAYOUT

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Nanometer CMOS VLSI, FinFET Physics & Integrated Circuit Layout.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Nanometer CMOS VLSI, FinFET Physics & Integrated Circuit Layout.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: FPGA ARCHITECTURE, HDL SYNTHESIS & STATIC TIMING ANALYSIS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of FPGA Architecture, HDL Synthesis & Static Timing Analysis.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of FPGA Architecture, HDL Synthesis & Static Timing Analysis.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: LOW-POWER EDGE COMPUTING, DSP & EMBEDDED TELEMETRY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Low-Power Edge Computing, DSP & Embedded Telemetry.
- [+] Synthesize and apply analytical frameworks across all 3 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Low-Power Edge Computing, DSP & Embedded Telemetry.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

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Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$

Equilibrium Balance Criterion: $\sum F_{net} = 0$, $\sum M$

Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \text{partial}$

Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES

Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v)$

Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$

Sample Variance Boundary: $s^2 = (1 / (n - 1)) * \text{sum}$

Curricular Efficiency Ratio: $\eta = (\text{Output_actual} /$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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